RESEARCH PORTFOLIO · 2026

Ifeoluwanimi P. Shobayo

BSc | MSPH Candidate, Epidemiology

Liberty University · Graduating May 2027

RESEARCH PORTFOLIO

Selected Research • Data Projects • Public Health Practice



EPIDEMIOLOGY • PUBLIC HEALTH RESEARCH • HEALTH DATA ANALYTICS

ipshobayo@gmail.com

ipshobayo.github.io/Research

IS

IFEOLUWANIMI P. SHOBAYO

MSPH CANDIDATE,
EPIDEMIOLOGY
LIBERTY UNIVERSITY

Contents

01 **Professional Profile** — Who I am and what I bring — p. 3

02 **Core Capabilities** — Six areas of practice — p. 4

03 **Professional Journey** — Research, clinical, and teaching roles — p. 5

04 **Portfolio at a Glance** — Sixteen projects in one table — p. 6

05 **Featured: Women's Mental Health Scoping Reviews** — MRKH and PMDD scoping reviews, first author — p. 7

06 **Featured: Environmental Health and Food Supply** — Campus green space and air quality (first author); HFCS in branded foods (co-author) — p. 8

07 **Featured: Perinatal Depression in Virginia** — CDC PRAMS trend analysis, VPHA 2026 — p. 9

08 **Featured: Mental Health Across Virginia Counties** — County-level analysis in Tableau, VPHA 2026 — p. 10

09 **Featured: Wearable Health Data** — Mixed-effects models on Fitbit data — p. 11

10 **Data Analysis Projects** — Framingham, food insecurity, machine learning, GIS, code — p. 12

11 **Evidence, Design & Policy** — Systematic review, study design, policy, program planning — p. 13

12 **Publications** — Peer-reviewed papers and conference posters — p. 14

13 **Skills & Education** — Methods, software, and degrees — p. 15

14 **Service & Recognition** — Abstract review, leadership, and community health — p. 16

15 **Research Direction & Contact** — Interests, goals, and how to reach me — p. 17

DATA · EVIDENCE · PREVENTION

IS

IFEOLUWANIMI P. SHOBAYO

MSPH CANDIDATE,
EPIDEMIOLOGY
LIBERTY UNIVERSITY

PORTFOLIO NAVIGATION

01 **Professional Profile**
02 Core Capabilities
03 Professional Journey
04 Portfolio at a Glance
05 Scoping Reviews
06 Air Quality & HFCS
07 Perinatal Depression
08 Virginia Mental Health
09 Wearable Health Data
10 Data Analysis Projects
11 Evidence, Design & Policy
12 Publications
13 Skills & Education
14 Service & Recognition
15 Research Direction & Contact

DATA · EVIDENCE · PREVENTION

01 | Professional Profile

Turning health data into clear answers

I am an MSPH Epidemiology candidate at Liberty University, graduating in May 2027, with a Bachelor of Science in Biological Sciences and hands-on experience in field surveys, laboratory research, and hospital infection prevention.

My research centers on women's and maternal mental health, infectious and chronic disease, and health equity. I have led six research posters as first author, including three accepted for the APHA 2026 Annual Meeting. I have also co-authored two more APHA posters (2025 and 2026) and four peer-reviewed publications.

I analyze cohort, surveillance, and wearable health data in SAS, R, and Tableau, build machine-learning models in Python, and map disease patterns in ArcGIS. I aim to report results clearly: what the data show, what they cannot show, and what they mean for public health practice.

My professional interests span epidemiology, public health research, health data analytics, digital health, and health data security, with an emphasis on using reliable data to inform public health decisions.

As a Graduate Student Assistant in Liberty's Department of Public Health, I teach class sessions, grade student assignments, and help prepare grant applications through the Office of Sponsored Programs & Research (OSPR).

6	4	4	4
first-author research posters	peer-reviewed publications	APHA 2026 posters, 3 as first author	APHA 2026 abstract-review programs

Seeking: 2027 epidemiology internships and entry-level roles in public health research and health data analytics.

02 | Core Capabilities

What I bring to an epidemiology or public health research team, drawn from the projects and roles in this portfolio.

EPIDEMIOLOGIC RESEARCH & SURVEILLANCE

Observational study design, research protocols and proposals, confounding and adjustment, and surveillance data such as CDC PRAMS and County Health Rankings.

BIOSTATISTICS & HEALTH DATA ANALYSIS

Linear, logistic, and mixed-effects regression; t-tests, chi-square tests, relative risks, and power calculations, with careful data cleaning.

VISUALIZATION & SCIENTIFIC COMMUNICATION

Tableau dashboards, county choropleth maps, ArcGIS heat maps, and research posters for APHA and VPHA audiences.

LITERATURE REVIEWS

Scoping, narrative, and systematic reviews (PRISMA-ScR, PRISMA): database searching, screening, data charting, and evidence mapping.

DATA MANAGEMENT & GOVERNANCE

Survey data collection with Kobo Collect, confidential data handling, credentialed-access wearable data, and cloud security hardening.

PROGRAM PLANNING & POLICY

PRECEDE needs assessment, behavioral theory and logic models, policy analysis, and grant application support through OSPR.

SOFTWARE SAS · R · SPSS · Excel · Python · SQL · Tableau · Power BI · ArcGIS · NVivo

IS

IFEOLUWANIMI P. SHOBAYO

MSPH CANDIDATE,
EPIDEMIOLOGY
LIBERTY UNIVERSITY

PORTFOLIO NAVIGATION

01 Professional Profile
02 Core Capabilities
03 Professional Journey
04 Portfolio at a Glance
05 Scoping Reviews
06 Air Quality & HFCS
07 Perinatal Depression
08 Virginia Mental Health
09 Wearable Health Data
10 Data Analysis Projects
11 Evidence, Design & Policy
12 Publications
13 Skills & Education
14 Service & Recognition
15 Research Direction & Contact

DATA · EVIDENCE · PREVENTION

03 | Professional Journey

Graduate Student Assistant
Liberty University, Department of Public Health
Aug 2026 – Present

Teach class sessions and grade student assignments in undergraduate public health courses, help prepare grant applications through Liberty's Office of Sponsored Programs & Research (OSPR), and support course instruction, Canvas course management, and students.

Clinical Observership: Infection Prevention, Quality & Patient Safety, Epidemiology
Emory Johns Creek Hospital, Emory Healthcare
Jul – Aug 2026

Observed CAUTI and CLABSI prevention rounds, case reviews (*C. difficile*, ESBL, VRE, TB), NHSN surveillance, and hand-hygiene quality work; wrote an SBAR recommendation on the electronic hand-hygiene monitoring workflow.

Biostatistics and Research Assistant
Liberty University, Department of Public Health
Jan – Aug 2026

Tutored students in biostatistics, supported faculty with data preparation and analysis, ran workshops, and conducted independent secondary analyses, including the wearable-data mixed-effects models.

Research Assistant
College of Medicine, Department of Public Health, University College Hospital, Ibadan
Jul – Nov 2024

Surveyed 100 workers across police stations in Ibadan for an occupational health study using Kobo Collect, safeguarded data accuracy and confidentiality, and helped present findings.

Research and Laboratory Assistant
Institute for Advanced Medical Research & Training (IAMRAT), UCH Ibadan
Apr – Oct 2021

Rotated through malaria, genetics, phytomedicine/virology, cancer, and neurobiobanking labs (microscopy, DNA/RNA extraction, PCR) and supported COVID-19 sample processing.

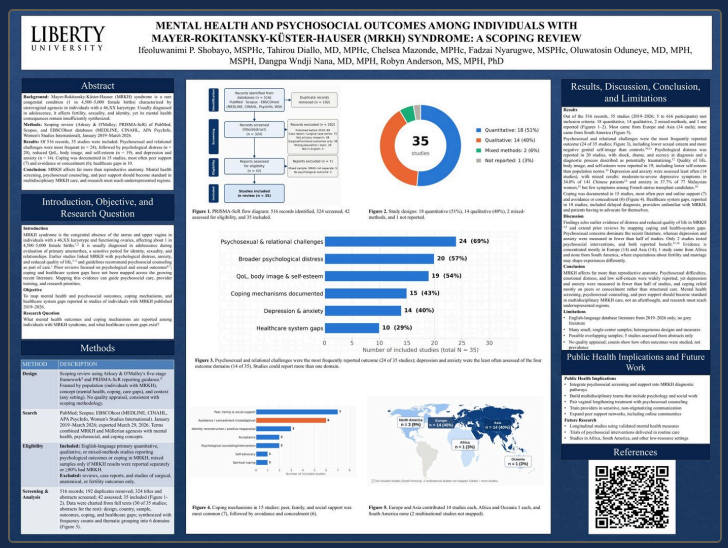
04 | Research Portfolio at a Glance

Sixteen projects across mental health, infectious and chronic disease, environmental health, health equity, and data science.

PROJECT	AREA	DESIGN & KEY RESULT	MY ROLE	OUTPUT
MRKH syndrome scoping review	Women's mental health	Scoping review, 35 studies included	First author	APHA 2026 poster
PMDD global scoping review	Women's mental health	Scoping review, 44 studies included	First author	APHA 2026 poster
Campus green space, air quality, and mental health	Environmental health	Narrative review, 15 studies included	First author	APHA 2026 poster
HFCS in U.S. branded food products	Nutrition surveillance	USDA database; 6.6% of eligible products listed HFCS	Co-author	APHA 2026 poster
Perinatal depression, Virginia	Maternal mental health	CDC PRAMS secondary trend analysis	First author	VPHA 2026 poster
Mental health across VA counties	Health equity	County Health Rankings, Tableau	First author	VPHA 2026 poster
Menstrual cycle, stress, and sleep	Digital health	Secondary analysis, mixed-effects models	Analyst	Case study
Framingham Heart Study	Cardiovascular disease	Secondary analysis, regression	Analyst	Case study
Food insecurity across U.S. states	Health equity	Secondary data, SAS, Tableau Prep	Team member	Group project
Predicting campaign responders	Machine learning	Python, SQL, cross-validation	Team member	Group project
1854 Soho cholera outbreak	Spatial epidemiology	ArcGIS Online heat map	Analyst	Map
Buruli ulcer therapy in Ghana	Infectious disease	Mini systematic review, PRISMA	Author	Review
Graduate student mental health	Study design	Cross-sectional design, DASS-21	Author	Proposal
California screening law	Health policy	Multiple streams analysis	Author	Policy analysis
Police hypertension, Ibadan	Program planning	PRECEDE, logic model	Author	Intervention proposal
IPC in long-term care facilities	Infection prevention	Mixed-methods protocol	Co-author	Manuscript in prep.

Two First-Author Scoping Reviews in Women's Mental Health

Arksey & O'Malley framework, reported with PRISMA-ScR · Liberty University research teams · Accepted posters

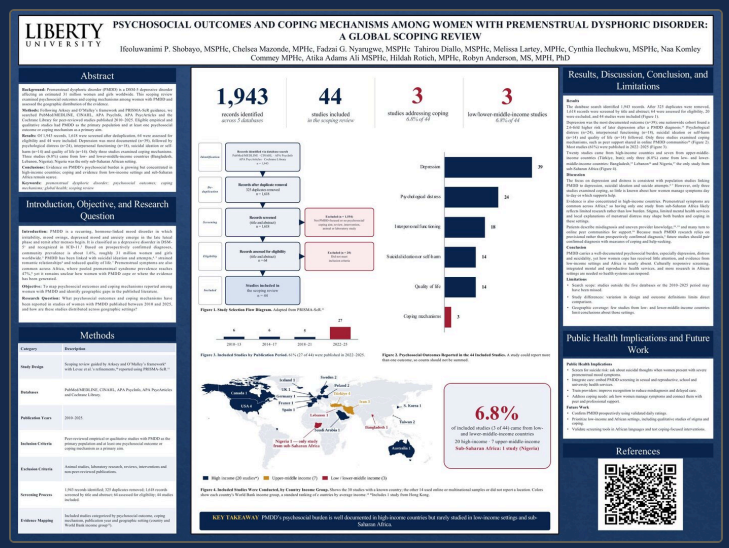


Mental Health and Psychosocial Outcomes in MRKH Syndrome

FIRST AUTHOR · 7 AUTHORS · 2019–2026 LITERATURE

516	324	42	35
RECORDS	SCREENED	ASSESSED	INCLUDED

- Outcomes reported: psychosexual and relational challenges (24 studies), psychological distress (20), quality of life, body image, and self-esteem (19), and depression and anxiety (14).
- Coping was documented in 15 studies, most often peer and online support (7) and avoidance or concealment (6); healthcare system gaps appeared in 10.
- Only 1 study came from Africa and none from South America.



Psychosocial Outcomes and Coping in PMDD: A Global Scoping Review

FIRST AUTHOR · 10 AUTHORS · 2010–2025 LITERATURE

1,943	1,618	64	44
RECORDS	SCREENED	ASSESSED	INCLUDED

- Outcomes reported: depression (39 studies), psychological distress (24), interpersonal functioning (18), suicidal ideation or self-harm (14), and quality of life (14).
- Only 3 studies examined coping mechanisms.
- Only 3 studies (6.8%) came from low- and lower-middle-income countries; Nigeria was the only sub-Saharan African setting.

IS

IFEOLUWANIMI P. SHOBAYO

MSPH CANDIDATE, EPIDEMIOLOGY
LIBERTY UNIVERSITY

PORTFOLIO NAVIGATION

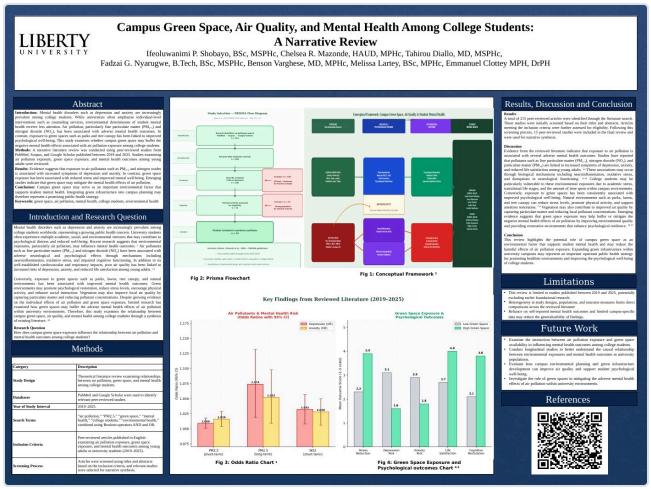
01 Professional Profile
02 Core Capabilities
03 Professional Journey
04 Portfolio at a Glance
05 Scoping Reviews
06 Air Quality & HFCS
07 Perinatal Depression
08 Virginia Mental Health
09 Wearable Health Data
10 Data Analysis Projects
11 Evidence, Design & Policy
12 Publications
13 Skills & Education
14 Service & Recognition
15 Research Direction & Contact

DATA · EVIDENCE · PREVENTION

06 · FEATURED RESEARCH

Environmental Health and Food Supply Research

Two posters accepted for APHA 2026: a narrative review as first author and a database analysis as co-author


Campus Green Space, Air Quality, and Mental Health Among College Students: A Narrative Review

LIBERTY UNIVERSITY

Ifeoluwanimi P. Shobayo, BSc, MSPHc, Chelsea R. Mazonde, HAUD, MPHc, Tahirou Diallo, MD, MSPHc, Fadzai G. Nyarugwe, B.Tech, BSc, MSPHc, Benson Varghese, MD, MPHc, Melissa Larrey, BSc, MPHc, Emmanuel Clottey MPH, DrPH

Abstract

Introduction: Mental health disorders such as depression and anxiety are increasingly prevalent among college students worldwide, representing a growing public health concern. University students frequently experience elevated psychological distress and reduced well-being. Recent research suggests that environmental exposures, particularly air pollution, may influence mental health outcomes.[1] Air pollutants such as particulate matter (PM) and nitrogen dioxide (NO_2) have been associated with adverse neurological and psychological effects through mechanisms including neuroinflammation, oxidative stress, and impaired cognitive functioning. In addition to its environmental benefits, urban green space has been consistently linked to improved mental health and reduced risk of psychiatric disorders among young adults.[2]

Objective: This narrative review synthesizes recent evidence on the relationship between air pollution and mental health among college students, and explores the potential role of campus green space as an environmental factor that may buffer or mitigate these effects.

Research Question: How does campus green space exposure influence the relationship between air pollution and mental health outcomes among college students?

Methods

Category	Description
Study Design	Theoretical literature review examining relationships between air pollution, green space, and mental health among college students.
Databases	PubMed and Google Scholar were used to identify relevant peer-reviewed studies.
Year of Study Interval	2014–2024
Search Terms	"air pollution," "PM2.5," "green space," "natural environment," "mental health," "depression," "anxiety," combined using Boolean operators AND and OR.
Inclusion Criteria	Peer-reviewed articles published in English examining air pollution exposure, green space exposure, and mental health outcomes among young adults or university students (2014–2024).
Screening Process	Articles were screened using titles and abstracts based on the inclusion criteria, and relevant studies were selected for narrative synthesis.

Fig 1: Conceptual Framework [3]

Fig 2: Prisma Flowchart

Key Findings from Reviewed Literature (2019–2025)

Fig 3: Odds Ratio Chart [4]

Fig 4: Green Space Exposure and Psychological outcomes Chart [5,6]

Results, Discussion and Conclusion

Limitations

Future Work

References

FIRST AUTHOR · 7 AUTHORS · NARRATIVE REVIEW

Campus Green Space, Air Quality, and Mental Health Among College Students

215	198	42	15
IDENTIFIED	SCREENED	ASSESSED	INCLUDED

- Exposure to $PM_{2.5}$ and NO_2 was associated with more symptoms of depression and anxiety and lower life satisfaction among young adults.
- Green space exposure was associated with reduced stress and better psychological well-being; emerging studies suggest it may buffer the mental health effects of air pollution.
- **Relevance:** campus green infrastructure as an upstream strategy for student mental health.


PREVALENCE OF HIGH-FRUCTOSE CORN SYRUP IN THE U.S. BRANDED FOOD SUPPLY: A CROSS-SECTIONAL ANALYSIS OF THE USDA GLOBAL BRANDED FOOD PRODUCTS DATABASE

LIBERTY UNIVERSITY

Mubaka F. Roberts, MD, MPH; Chelsea R. Mazonde, MPHc; Ifeoluwanimi P. Shobayo, MSPHc

Abstract

Introduction, Objective, and Research Question

Methods

Results, Discussion, Conclusion, and Limitations

Public Health Implications and Future Work

References

CO-AUTHOR · 3 AUTHORS · CROSS-SECTIONAL ANALYSIS

High-Fructose Corn Syrup in the U.S. Branded Food Supply

454,324	447,607	29,503
IDENTIFIED	ELIGIBLE	LIST HFCS

- Retrospective, cross-sectional, descriptive analysis of the USDA Global Branded Food Products Database (products published 2019–2025).
- HFCS was listed in 6.6% of eligible products, about 1 in 15. Counts are not sales-weighted, so this is a product-level baseline, not a measure of intake.
- **Relevance:** a reproducible baseline for monitoring HFCS, added sugars, and reformulation.

Both are secondary research: a review of published studies and an analysis of an existing public database. Neither involved primary data collection.

IS

IFEOLUWANIMI P. SHOBAYO

MSPH CANDIDATE, EPIDEMIOLOGY
LIBERTY UNIVERSITY

PORTFOLIO NAVIGATION

01 Professional Profile
02 Core Capabilities
03 Professional Journey
04 Portfolio at a Glance
05 Scoping Reviews
06 Air Quality & HFCS
07 Perinatal Depression
08 Virginia Mental Health
09 Wearable Health Data
10 Data Analysis Projects
11 Evidence, Design & Policy
12 Publications
13 Skills & Education
14 Service & Recognition
15 Research Direction & Contact

DATA · EVIDENCE · PREVENTION

07 · FEATURED ANALYSIS

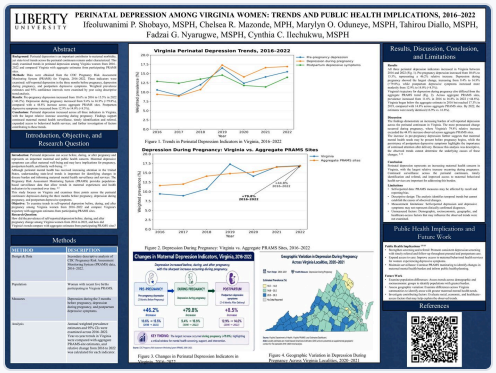
Perinatal Depression Among Virginia Women, 2016–2022

CDC PRAMS surveillance · Virginia Public Health Association 2026 Annual Conference

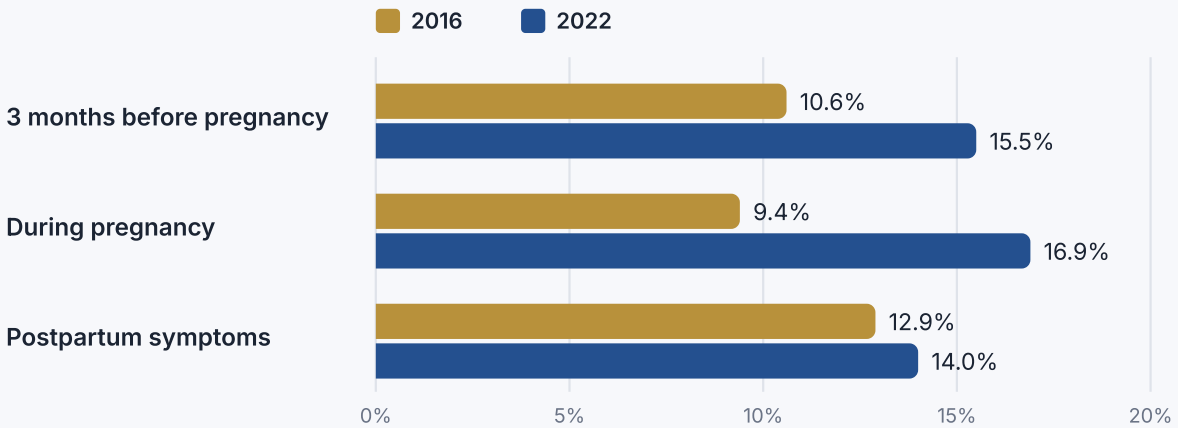
ROLE	DATA	DESIGN	MEASURES
First author	CDC PRAMS, Virginia, 2016–2022	Secondary descriptive trend analysis	Weighted prevalence, 95% CIs

+79.8%

relative rise in depression during pregnancy in Virginia (9.4% to 16.9%), compared with +44.8% (11.6% to 16.8%) across aggregate PRAMS sites.



SELF-REPORTED DEPRESSION, VIRGINIA, 2016 VS. 2022



PUBLIC HEALTH IMPLICATIONS

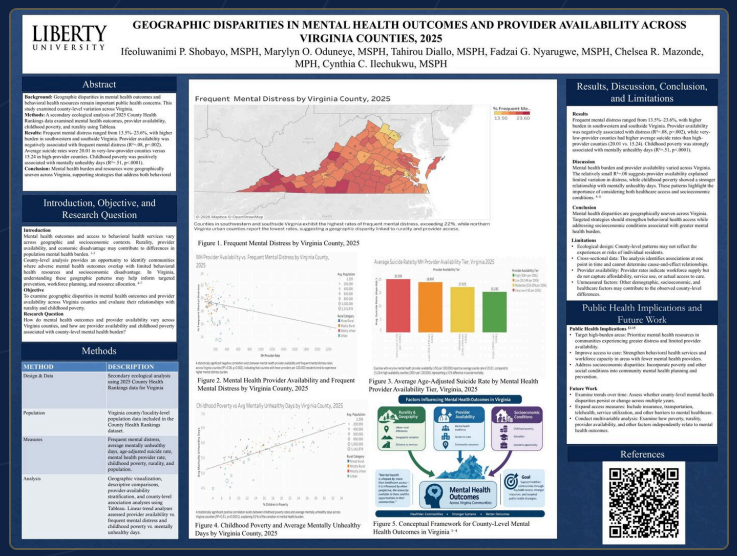
- Consistent depression screening with referral and follow-up through prenatal and postpartum care.
- Better access to maternal behavioral health services.
- Continued PRAMS surveillance, analyzed by population group and locality.

LIMITATIONS

Annual estimates fluctuated (each indicator peaked in 2019 and dropped in 2020), and the small postpartum change may be within sampling error. The estimates are self-reported, no trend tests were reported, and a descriptive design cannot explain why rates rose.

Geographic Disparities in Mental Health Across Virginia Counties

Secondary ecological analysis in Tableau · 2025 County Health Rankings · First author, Virginia Public Health Association 2026


LIBERTY UNIVERSITY

GEOGRAPHIC DISPARITIES IN MENTAL HEALTH OUTCOMES AND PROVIDER AVAILABILITY ACROSS VIRGINIA COUNTIES, 2025

Ifeoluwanimi P. Shobayo, MSPH, Marylyn O. Oduneye, MSPH, Tahirou Diallo, MSPH, Fadzai G. Nyarugwe, MSPH, Chelsea R. Mazonde, MPH, Cynthia C. Ilechukwu, MSPH

Abstract

Introduction, Objective, and Research Question

Methods

Frequent Mental Distress by Virginia County, 2025

Figure 1. Frequent Mental Distress by Virginia County, 2025

Figure 2. Mental Health Provider Availability and Frequent Mental Distress by Virginia County, 2025

Figure 3. Average Age-Adjusted Suicide Rate by Mental Health Provider Availability Tier, Virginia, 2025

Figure 4. Childhood Poverty and Average Mentally Unhealthy Days by Virginia County, 2025

Figure 5. Conceptual Framework for County-Level Mental Health Outcomes in Virginia

Results, Discussion, Conclusion, and Limitations

Public Health Implications and Future Work

References

METHODS AND MY ROLE

- County measures: frequent mental distress, mentally unhealthy days, suicide rate, mental health providers, and childhood poverty.
- County choropleth map, four provider-availability tiers, and linear trend analyses in Tableau.
- Group poster with six authors. I am first author, and its topic, dataset, and study questions follow my individual project proposal (January 2026).

13.5–23.6%

frequent mental distress across counties, highest in southwestern and southside Virginia

$R^2 = 0.51$

childhood poverty vs. mentally unhealthy days ($p < 0.0001$)

$R^2 = 0.08$

provider supply vs. frequent mental distress (negative, $p = 0.002$)

20.0 vs. 15.2

average suicide rate per 100,000, lowest vs. highest provider tier (descriptive)

WHAT IT MEANS

Childhood poverty tracked county mental health burden far more closely than provider supply did. These are unadjusted, county-level associations, so they show where burden, poverty, and provider supply overlap rather than what would reduce distress. Together they support attention to both behavioral health access and socioeconomic conditions.

Tableau · Choropleth mapping · Ecological analysis · Health equity

IS

IFEOLUWANIMI P. SHOBAYO

MSPH CANDIDATE, EPIDEMIOLOGY
LIBERTY UNIVERSITY

PORTFOLIO NAVIGATION

01 Professional Profile
02 Core Capabilities
03 Professional Journey
04 Portfolio at a Glance
05 Scoping Reviews
06 Air Quality & HFCS
07 Perinatal Depression
08 Virginia Mental Health
09 **Wearable Health Data**
10 Data Analysis Projects
11 Evidence, Design & Policy
12 Publications
13 Skills & Education
14 Service & Recognition
15 Research Direction & Contact

09 · FEATURED ANALYSIS

Menstrual Cycle, Stress, and Sleep in Wearable Data

Secondary analysis of the mcPHASES dataset (PhysioNet, 42 participants) · Linear mixed-effects regression · April 2026

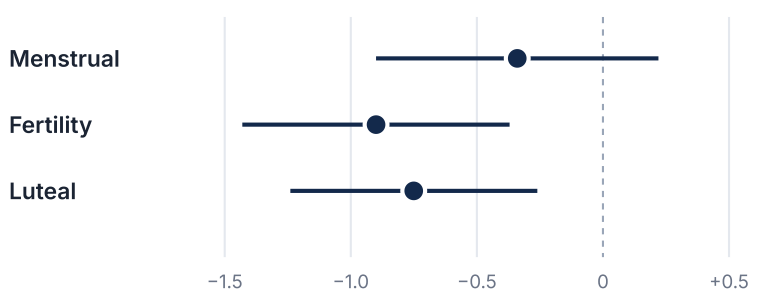
BUILDING THE ANALYTIC DATASET

1. Merged activity, sleep, stress, hormone-phase, and participant files on ID, interval, and study day
2. Removed duplicates, keeping one record per person-day
3. Linked each day's activity to the following night's sleep score
4. Kept valid stress readings and derived age at each study interval
5. Excluded days missing the outcome, exposure, phase, or age

Analysis 1 · Menstrual phase and daily stress score

3,732 PERSON-DAYS · 36 PEOPLE

Adjusted difference in the Fitbit Stress Management Score vs. the follicular phase (below 0 = more physical signs of stress), adjusted for age, weekend, and study period.



	β (95% CI)	p
Menstrual	−0.34 (−0.90 to 0.22)	0.23
Fertility	−0.90 (−1.43 to −0.37)	<0.001
Luteal	−0.75 (−1.24 to −0.26)	0.003

Fertility and luteal days showed more physical signs of stress than follicular days. In a sensitivity analysis, only the fertility-phase difference held, making it the most stable finding.

Analysis 2 · Physical activity and next-night sleep

4,893 PERSON-DAYS · 42 PEOPLE

Each extra hour of activity was associated with a sleep score only about 0.04 points higher (95% CI −0.07 to 0.15; p = 0.46): no meaningful association. Menstrual phase was associated with sleep score (p < 0.001).

Why mixed models: days from the same person are correlated, so ordinary regression would understate uncertainty. A random intercept for each participant accounts for that clustering.

10 | Data Analysis Projects

Secondary analyses of existing cohort, surveillance, and administrative datasets in SAS, R, Python, Excel, Tableau, and ArcGIS.

FRAMINGHAM HEART STUDY

4,434 PARTICIPANTS

Secondary analysis of the NHLBI teaching dataset. Diabetes' link to heart rate (+4.3 beats/min) shrank to +0.9 and lost significance after adjustment, largely absorbed by glucose.

t-tests · Regression · Excel

FOOD INSECURITY ACROSS U.S. STATES

255 STATE-YEAR RECORDS

Group project: secondary analysis of Feeding America data, 2019–2023. Rates dipped during pandemic relief, then climbed to an average state rate of 14.2% by 2023, highest in the South.

SAS · Tableau Prep

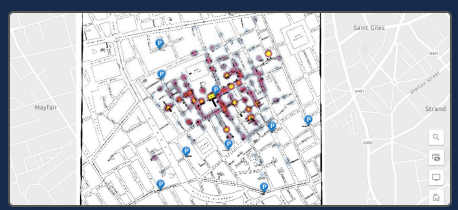
PREDICTING CAMPAIGN RESPONDERS

11,162 RECORDS

Group project on a public dataset: K-Means segmentation and four classifiers with cross-validation, keeping a leaking variable (call duration) out of the model.

Python · SQL · Machine learning

MAPPING THE 1854 SOHO CHOLERA OUTBREAK



An ArcGIS Online heat map recreating John Snow's investigation from historical data, plus reviews of spatial clustering and dashboard design.

ArcGIS · Spatial epidemiology

STATISTICAL PROGRAMMING

3 LANGUAGES

SAS assumption checks, regression, chi-square relative risks, and power calculations; a tidyverse cleaning script in R; and a pandas data-quality diagnosis in Python.

SAS · R · Python

FULL CASE STUDIES ONLINE

16 PROJECT PAGES

Most projects in this portfolio have a full online case study with methods, results tables, and limitations at ipshobayo.github.io/Research.

Portfolio website

IS

IFEOLUWANIMI P. SHOBAYO

MSPH CANDIDATE,
EPIDEMIOLOGY
LIBERTY UNIVERSITY

PORTFOLIO NAVIGATION

01 Professional Profile
02 Core Capabilities
03 Professional Journey
04 Portfolio at a Glance
05 Scoping Reviews
06 Air Quality & HFCS
07 Perinatal Depression
08 Virginia Mental Health
09 Wearable Health Data
10 Data Analysis Projects
11 **Evidence, Design & Policy**
12 Publications
13 Skills & Education
14 Service & Recognition
15 Research Direction & Contact

DATA · EVIDENCE · PREVENTION

11 | Evidence, Design & Policy

MINI SYSTEMATIC REVIEW OF PUBLISHED STUDIES · PRISMA

Antibiotic Therapy for Buruli Ulcer in Ghana

Searched six databases, identified 126 records, and synthesized nine sources. Oral rifampicin–clarithromycin healed over 90% of mostly small lesions in the included studies, while resistance monitoring remains a need.

CROSS-SECTIONAL STUDY DESIGN (PROPOSAL)

Stress, Anxiety, and Social Support Among Graduate Students

Designed a study comparing international and domestic graduate students, with sample-size planning, validated instruments (DASS-21, MSPSS), confounders, and a regression and moderation analysis plan.

POLICY ANALYSIS

California's Maternal Mental Health Screening Law

Applied Kingdon's multiple streams framework. Screening rose from 51.4% to 65.9% of mothers (2020–2024), but only 47.1% of those with symptoms received treatment, so the gap between screening and care is the law's main weakness.

INTERVENTION PROPOSAL

Reducing Stress-Related Hypertension Among Police Officers in Ibadan

A workplace program built on a PRECEDE needs assessment, Social Cognitive Theory and the IMB model, a logic model, a stakeholder coalition, and a pre/post evaluation plan.

RESEARCH PROTOCOL · CO-AUTHOR · MANUSCRIPT IN PREPARATION

Infection Prevention and Control in Long-Term Care Facilities

A mixed-methods pilot feasibility study of IPC infrastructure vulnerability in long-term care facilities following Epidemiology and Laboratory Capacity (ELC) funding withdrawal.

12 | Publications & Presentations

4	6	4	1
peer-reviewed publications	first-author research posters	APHA 2026 posters, 3 as first author	manuscript in preparation

PEER-REVIEWED PUBLICATIONS

Linus EN, Anih DC, Iwuajoku D, Apata CO, **Shobayo IP**, Chindo AM. Vaccine hesitancy and immunization inequity in Nigeria: public health determinants, misinformation dynamics, and health system strategies for improved uptake. *Int J Health Pharm Res*. 2026;11(8):32-56.

Edeh CG, Darkoh GO, **Shobayo IP**, et al. A machine learning-based framework for real-time environmental health risk prediction and spatial air quality intelligence in urban-industrial ecosystems. *Asian J Adv Res Rep*. 2026;20(4):1-20.

Oladokun VEE, Brown JO, **Shobayo IP**, Zainab GA, Chinonyerem CA. Integrating sensory evaluation and food safety assessment to improve consumer acceptance and public health outcomes of ready-to-eat snacks. *Asian J Adv Res Rep*. 2026;20(4):268-283.

Odekunle YO, **Shobayo IP**, Edobor O, et al. Modelling and mitigating antimicrobial resistance (AMR) through data-driven surveillance, AI-powered drug discovery, and public health intervention design. *Middle East J Appl Innov Multidiscip Res*. 2025;9(2).

MANUSCRIPT IN PREPARATION

Okojie PW, Anderson R, Arsin KJ, Sarpong PO, **Shobayo I**. Protocol for mixed-methods pilot feasibility study of infection prevention and control infrastructure vulnerability in long-term care facilities following ELC funding withdrawal.

OTHER CONFERENCE PRESENTATIONS

VPHA 2026 Annual Conference
Shobayo IP, et al. Geographic disparities in mental health outcomes and provider availability across Virginia counties, 2025.

APHA 2025 Annual Meeting
Agbotui B, **Shobayo I**, Anderson R. E-cigarette use and depression among adults in Virginia: a cross-sectional analysis of 2023 BRFSS data.

SELECTED CONFERENCE PRESENTATIONS: APHA 2026

First author
Shobayo IP, Diallo T, Mazonde C, Nyarugwe F, Oduneye O, Wndji Nana D, Anderson R. Mental health and psychosocial outcomes among individuals with Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome: a scoping review.

First author
Shobayo IP, Mazonde C, Nyarugwe FG, Diallo T, Lartey M, Ilechukwu C, Commey NK, Adams Ali A, Rotich H, Anderson R. Psychosocial outcomes and coping mechanisms among women with premenstrual dysphoric disorder: a global scoping review.

First author
Shobayo IP, Mazonde CR, Diallo T, Nyarugwe FG, Varghese B, Lartey M, Clottey E. Campus green space, air quality, and mental health among college students: a narrative review.

Co-author
Roberts MF, Mazonde CR, **Shobayo IP**. Prevalence of high-fructose corn syrup in the U.S. branded food supply: a cross-sectional analysis of the USDA Global Branded Food Products Database.

VPHA 2026 Annual Conference
Shobayo IP, et al. Perinatal depression among Virginia women: trends and public health implications, 2016–2022.

Liberty University Research Week 2026
Shobayo I, et al. Mpox in non-MSM populations (systematic review proposal) and campus green space, air quality, and mental health; co-author on 8 more posters.

IS

IFEOLUWANIMI P. SHOBAYO

MSPH CANDIDATE, EPIDEMIOLOGY
LIBERTY UNIVERSITY

PORTFOLIO NAVIGATION

01 Professional Profile
02 Core Capabilities
03 Professional Journey
04 Portfolio at a Glance
05 Scoping Reviews
06 Air Quality & HFCS
07 Perinatal Depression
08 Virginia Mental Health
09 Wearable Health Data
10 Data Analysis Projects
11 Evidence, Design & Policy
12 Publications
13 **Skills & Education**
14 Service & Recognition
15 Research Direction & Contact

DATA · EVIDENCE · PREVENTION

13 | Skills & Education

EPIDEMIOLOGIC RESEARCH & SURVEILLANCE

- Observational study design (cohort, cross-sectional)
- Research protocols and grant application preparation
- Surveillance data (CDC PRAMS, County Health Rankings)
- Program planning and evaluation (PRECEDE, logic models)
- Laboratory methods: microscopy, DNA/RNA extraction, PCR

BIOSTATISTICS & HEALTH DATA ANALYSIS

- Linear, logistic, and mixed-effects regression
- t-tests, chi-square tests, relative risks, and correlation
- Confounding and adjustment
- Power and sample size calculations
- Machine learning: clustering and classification

SCOPING & NARRATIVE LITERATURE REVIEWS

- Scoping reviews (Arksey & O'Malley, PRISMA-ScR)
- Narrative reviews
- Systematic review methods (PRISMA)
- Reference management: Zotero, Mendeley, RefWorks

RESEARCH DATA MANAGEMENT

- Survey data collection (Kobo Collect)
- Data cleaning, merging, and quality checks
- Linking multi-file datasets (person-day records)
- Documentation of exclusions and analytic samples

DATA VISUALIZATION & SCIENTIFIC COMMUNICATION

- Tableau dashboards and county choropleth maps
- ArcGIS heat maps and spatial epidemiology
- Scientific posters (APHA, VPHA)
- Teaching undergraduate class sessions

DIGITAL HEALTH & HEALTH DATA GOVERNANCE

- Wearable (Fitbit) health data analysis
- Credentialed-access data under a data use agreement (PhysioNet)
- Confidential data handling and data privacy
- Cybersecurity: cloud hardening (CyberGirls)

SOFTWARE SAS · R · SPSS · Excel · Python · SQL · Tableau · Power BI · ArcGIS · NVivo

EDUCATION

Master of Science in Public Health (MSPH), Epidemiology

Liberty University, Lynchburg, VA

Anticipated May 2027 · GPA 4.0

Coursework: biostatistics, epidemiologic research methods, SAS, field epidemiology (R), data visualization and spatial analysis, chronic and infectious disease.

Bachelor of Science, Biological Sciences

Federal University of Agriculture, Abeokuta, Nigeria

March 2023

Thesis: neurotransmitter (dopamine and serotonin) activity in Wistar rats exposed to ultraviolet and infrared light.

14 | Service, Leadership & Recognition

AMERICAN PUBLIC HEALTH ASSOCIATION · 2026

Abstract Reviewer, APHA 2026 Annual Meeting

Recognized with certificates of appreciation for reviewing abstracts for four programs:

- Epidemiology
- Epidemiology Late Breakers
- Occupational Health and Safety Late Breakers
- Community Health Planning and Policy Development Late Breakers

ASPIRE INSTITUTE · SEPTEMBER 2026

Aspire Leaders Program

Selected for the fully funded global leadership program and completed its modules, including a strengths-based leadership assessment and the Aspire Horizons course on leadership and community impact.

MEMBERSHIPS

- American Public Health Association (APHA)
- Virginia Public Health Association
- Liberty University Public Health Student Association
- Pure and Applied Zoology Students' Association

250+

people reached through genotype screening and awareness campaigns

COMMUNITY HEALTH

- **Public Health Outreach Leader**, Sickle Cell Hope Alive Foundation, Ibadan (2021 – present): genotype screening and awareness campaigns.
- **Volunteer**, Virginia Department of Health, Near Southwest Medical Reserve Corps (2025 – present): supporting the transition to electronic medical records in the Central Virginia Health District.
- **Medical outreach volunteer**, WOCOM, Ibadan (2018 – 2024): blood pressure screening and anthropometric measurements in underserved communities.
- Co-organizer, LUPHSA Health Fair (2025, 2026) · AHEV leadership member · Out of the Darkness Walk team.

IS

IFEOLUWANIMI P. SHOBAYO

MSPH CANDIDATE,
EPIDEMIOLOGY
LIBERTY UNIVERSITY

PORTFOLIO NAVIGATION

01 Professional Profile
02 Core Capabilities
03 Professional Journey
04 Portfolio at a Glance
05 Scoping Reviews
06 Air Quality & HFCS
07 Perinatal Depression
08 Virginia Mental Health
09 Wearable Health Data
10 Data Analysis Projects
11 Evidence, Design & Policy
12 Publications
13 Skills & Education
14 Service & Recognition
15 **Research Direction & Contact**

DATA · EVIDENCE · PREVENTION

15 | Research Direction & Contact

I am building toward a career in applied epidemiology and public health research, and I am seeking 2027 epidemiology internships and entry-level roles where I can put these skills to work.

WOMEN'S & MATERNAL MENTAL HEALTH

Scoping reviews on MRKH and PMDD, perinatal depression trends in CDC PRAMS, and screening policy.

INFECTIOUS DISEASE & INFECTION PREVENTION

Hospital infection-prevention exposure, an IPC research protocol, Buruli ulcer evidence, and laboratory work.

CHRONIC DISEASE & NUTRITION

Framingham risk-factor analysis, a workplace hypertension intervention, and HFCS food supply surveillance.

ENVIRONMENTAL HEALTH & HEALTH EQUITY

Campus green space and air quality, county-level disparities in Virginia, and food insecurity.

HEALTH DATA ANALYTICS

Mixed-effects models, machine learning, Tableau dashboards, and GIS mapping.

DIGITAL HEALTH & HEALTH DATA SECURITY

Wearable health data under a data use agreement, confidential data handling, and cloud security hardening.

LET'S CONNECT

Ifeoluwanimi P. Shobayo

MSPH Candidate, Epidemiology
Liberty University

ipshobayo@gmail.com

LinkedIn profile: linkedin.com/in/ifeoluwanimi-shobayo-358a441b3

ipshobayo.github.io/Research

Scan to save my contact, download my CV, or view my research.